

SEM Project Proposal

Project Description:

Knowing alloy composition is required for replicability and manufacturing of novel alloys. To repeat past experiments and reproduce results, manufacturers need to know what the final composition of the alloy is as it can change between the weighed powder composition and manufacturing. Inductively Coupled Plasma Mass Spectrometry (ICP-MS) is a very accurate method for determining composition but is time intensive and destroys the sample in the process. Energy-Dispersive X-ray Spectroscopy (EDX) on the other hand conserves the sample and is less time intensive but loses compositional accuracy. EDX, if optimized properly, offers a better alternative to ICP-MS for determining sample composition. Our project is to optimize the microscope parameters for EDX to minimize the error between EDX and ICP-MS composition readings using Bayesian Optimization, a machine learning technique for better experimental design.

Sample description:

The samples are polished portions of AlSi10Mg powder manufactured using Laser Powder Bed Fusion (LPBF). They are electrically conductive. Using both secondary electron imaging, we will select a clean spot from where we will use EDX to collect data for elemental analysis of our alloy.

Week	Dates	Completed Tasks
1	Sep 3 - Sep 5	First Draft of Proposal
2	Sep 6 - Sep 12	Finish SEM Training
3	Sep 13 - Sep 19	Have completed SEM pictures on first sample, prepare more samples in polishing labs
4	Sep 20 - Sep 26	Train on EDX
5	Sep 27 - Oct 3	Get EDX data for samples 1-6
6	Oct 4 - Oct 10 (PR#1)	Get EDX data for samples 7-12
7	Oct 11 - Oct 17	Get EDX data for samples 13-18
8	Oct 18 - Oct 24	Polish new samples
9	Oct 25 - Oct 31	Use optimization method for new samples 1-4

10	Nov 1 - Nov 7	Use optimization method for new samples 5-8
11	Nov 8 - Nov 14	Use optimization method for new samples 9-12
12	Nov 15 - Nov 21 (PR#2)	Use optimization method for new samples 13-16, begin write up
13	Nov 22 - Nov 28	Finish write up and presentation